

Test Yourself

Microeconomics

Cross Elasticity of

Demand

Section A

Duration of each question is **40 mins**

1. With the help of a formula, explain the meaning of cross elasticity of demand and consider which determinants are most important in establishing the size and sign of its coefficient. (8) **q3a/21/M/J/23**
2. With the help of a diagram(s), briefly explain how the cause of a movement along the demand curve differs from the cause of a shift in the demand curve and consider the impact on the demand for one good of price changes in other goods. (8) **q2a/22/O/N/24**

Section B

Duration 40 min

1. Following the introduction of a 10% increase in charges for car parking in a city centre, the demand for bus journeys per day into the city rose from 800 to 1000.

What was the cross elasticity of demand for bus journeys with respect to car parking charges?

- A -0.4 B +0.4 C +2.0 D +2.5

2. The table gives estimates of the price elasticities and cross elasticities of demand for bus and rail travel.

service	elasticity with respect to the price of	
	bus travel	rail travel
bus travel	-0.37	+0.13
rail travel	+0.16	-0.43

What would be the change in the volume of rail travel resulting from a 1% increase in bus fares?

- A a decrease of 0.13%
B a decrease of 0.37%
C an increase of 0.16%
D an increase of 0.43%
3. A government introduces a subsidy for product X and the market price of X falls by 5%. The cross elasticity demand for product Y, with respect to the price of product X, is +0.4.

What will happen to the demand for product Y?

- A It will decrease by 2%.
B It will decrease by 12.5%.
C It will increase by 2%.
D It will increase by 12.5%.

4. Knowledge of which combination of demand elasticities would be relevant to a firm trying to increase its revenue?

	price elasticity of demand	cross elasticity of demand	income elasticity of demand
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

5. The table gives information about the market for two models of car.

model	number of cars sold per week	cross elasticity of demand with respect to the price of petrol
1200 cc	10 000	-0.25
2000 cc	5 000	-0.50

If the prices of the cars remain unchanged, but the price of petrol increases by 100%, what will be the effect on the number of cars sold per week?

- A** increase by 5000
 - B** no change
 - C** decrease by 5000
 - D** decrease by 15 000
6. What can be measured by cross-elasticity of demand?
- A** a change in real income as a result of a change in the price of consumer goods
 - B** a change in the demand for a good in response to a change in the price of a complement
 - C** a change in the price of a good in response to a change in the demand for a substitute
 - D** a change in the price of a good when the demand for it changes

7. The table shows the quantity demanded of three goods when the price of good X changes.

price of X (\$)	quantity of X demanded	quantity of Y demanded	quantity of Z demanded
5	100	60	20
4	110	50	22
3	140	30	28
2	200	15	40
1	300	4	60

Which relationships between X, Y and Z can be deduced from the table?

	Y is	Z is
A	a complement for X	a complement for X
B	a complement for X	a substitute for X
C	a substitute for X	a complement for X
D	a substitute for X	a substitute for X

8. The cross-elasticity of demand of good S with respect to the price of good P is +1.5.
The cross-elasticity of demand of good S with respect to the price of good R is -1.5.
The cross-elasticity of demand of good P with respect to the price of good R is -1.5.

What can be concluded about goods P, R and S?

- A** S and P are complements; P and R are substitutes.
- B** S and P are complements; R is an inferior good.
- C** S and P are substitutes; P and R are complements.
- D** S and P are substitutes; R is an inferior good.

9. What can be concluded about a good with a positive cross-price elasticity of demand?

- A** Its price will be sensitive to changes in prices of close substitutes.
- B** Its price will be sensitive to changes in quantity demanded of close substitutes.
- C** The quantity demanded for this good will be sensitive to changes in prices of close substitutes.
- D** The quantity demanded for this good will be sensitive to changes in quantity demanded of close substitutes.

10. Which row correctly describes the relationship between two goods implied by the value of the cross elasticity of demand?

	value of the cross elasticity of demand	description
A	-2.0	weak complements
B	-0.5	weak substitutes
C	+0.5	weak substitutes
D	+2.0	strong complements

11. The cross elasticity of demand between bus travel and rail travel is +2. A rise in the price of bus fares caused the demand for rail travel to rise by 10%.

Which change in bus fares would have caused this?

	from (cents)	to (cents)
A	20	21
B	30	33
C	40	45
D	50	60

12. The cross-elasticity of demand between good P and good Q is -0.4.

The cross-elasticity of demand between good P and good R is +0.6.

Which combination of goods is most likely to be represented by this information?

	good P	good Q	good R
A	cream	yoghurt	raspberry
B	yoghurt	strawberry	raspberry
C	strawberry	cream	raspberry
D	strawberry	raspberry	cream

13. In a town, bus fares fall by 50% and this leads to an increase in bus use by 30%. The effect on car use is very low with only a 1% reduction.

What is the value of the cross-elasticity of demand between car travel and bus travel?

- A** -0.60 **B** -0.02 **C** +0.02 **D** +0.60

14. A good has the following elasticity values.

cross elasticity of demand	+1.4
income elasticity of demand	-0.8

Which statement is correct?

- A** The good is a complement and inferior.
B The good is a complement and normal.
C The good is a substitute and inferior.
D The good is a substitute and normal.

15. The price elasticity of demand for good X is -2.4, its income elasticity of demand is -0.4 and the cross elasticity of demand for good X with respect to good Y is +0.8.

What is the correct description of good X?

- A** inferior good, price-elastic demand and substitute for good Y
B inferior good, price-inelastic demand and complement to good Y
C normal good, price-elastic demand and complement to good Y
D normal good, price-inelastic demand and substitute for good Y

16. Good X is related to good Y. Incomes increase and the price of good Y falls.

Which combination would give the greatest increase in demand for good X?

	nature of good X	relationship between good X and good Y
A	inferior good	complements
B	inferior good	substitutes
C	normal good	complements
D	normal good	substitutes

17. The cross elasticity of demand between two products, X and Y, is negative.

What would be the immediate effect of a rise in the price of product Y?

- A Quantity demanded of product X will fall.
- B Supply of product X will rise.
- C The cross elasticity of demand will rise.
- D The price of product X will rise.

18. Two goods, X and Y, have a cross elasticity of demand of +1.8. Good Y has an income elasticity of demand of -0.6.

An increase in both the incomes of consumers and the price of good Y will have which combination of effects?

	quantity demanded of X	quantity demanded of Y
A	fall	fall
B	fall	rise
C	rise	fall
D	rise	rise

19. Travellers can cross the English Channel either by the Channel Tunnel or by a sea journey.

The table gives details of the passengers, in thousands, using these services between 2005 and 2015.

Between 2005 and 2015 the price of the Channel Tunnel fell and the price of sea journeys rose.

year	Channel Tunnel passengers	sea journey passengers
2005	17 005	23 693
2010	18 303	21 883
2015	20 864	21 000

What does the data suggest?

- A The cross elasticity of demand between the two services is positive.
- B The market for crossing the English Channel grew continuously over the period.
- C The rate of decline in sea journeys accelerated over the period.
- D The two services for crossing the English Channel are complementary.

20. The table gives a set of price and cross-price elasticities for four commodities, W, X, Y and Z.

with respect to price of	elasticity of demand for commodity			
	W	X	Y	Z
W	−0.3	−0.5	+0.7	−1.3
X	−0.7	−1.2	+0.8	+1.6
Y	+1.3	+1.7	−3.6	+0.8
Z	−2.7	+2.6	+1.7	−0.7

Which pairs of commodities are complements?

- A W and X; W and Z
- B W and Y; X and Y
- C W and Z; X and Z
- D X and Y; X and Z

MARKING SCHEME SECTION A

Question	Answer	Marks
1	With the help of a formula, explain the meaning of cross elasticity of demand <u>and</u> consider which determinants are most important in establishing the size and sign of its coefficient. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and understanding (3 marks maximum) An understanding of what the cross elasticity of demand measures (1), the formula for cross elasticity of demand (1) and the distinction between positive and negative XED. (1) AO2 Analysis (3 marks maximum) Explanation of why XED can be positive (substitutes) (1), explanation of why XED can be negative (complements) (1) and explanation of determinants establishing its magnitude (the closer the relationship between two products, whether substitutes or complements, the greater the magnitude of the value of XED). (1) AO3 Evaluation (max 2 marks) Offers a valid judgement on the relative importance of the determinants that influence the magnitude of XED and whether it is positive or negative (1) to reach a conclusion. (1)	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
2	With the help of a diagram(s), briefly explain how the cause of a movement along the demand curve differs from the cause of a shift in the demand curve <u>and</u> consider the impact on the demand for one good of price changes in other goods. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding, up to 3 marks for AO2 Analysis, and up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) • An accurate demand (and supply) diagram/or 2 separate diagrams clearly demonstrating a movement and a shift (1). NB: it is not essential to show the supply curve or changing price and output. • Explanation that a movement occurs due to a change in price (1) and that a shift occurs due to a change in a non-price factor (1). AO2 Analysis (max 3 marks) Candidates may analyse the impact on the demand for one good as a result of whether they are substitutes or complements of other goods (of any valid relationships between goods): This will include an analysis of the impact on demand for one good because of e.g.: • Price changes and the availability of complements (a negative value indicates a complement) e.g. a rise in the price of the complement would be expected to reduce the demand for the 'other good'. • Price changes and the availability of substitutes (a positive value indicates a substitute) e.g. a rise in the price of the substitute would be expected to increase the demand for the 'other good'. • For example, an increase in the price of a complement good is likely to reduce the demand of the good (1) due to the nature of the joint demand between both goods (1). • For example, a rise in the price of a substitute good is likely to increase the demand for the other good (1) if the consumer now sees the other good as a direct replacement (1). • Note: Diagrams are not essential. Note: Maximum of 2 marks if only one relationship is analysed. References to changes in income/fashion/tastes/attitudes must not be rewarded.	8

Question	Answer	Marks
2	AO3 Evaluation (max 2 marks) Up to 2 marks for considering the overall impact of these two factors depending on the size and values of the XED coefficients on the demand for one product (1) leading to a valid conclusion about the overall size of the impact of changes in the price of 'other goods (1) .	
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

SECTION B

1.	D	8.	C	15.	A
2.	C	9.	C	16.	C
3.	A	10.	C	17.	A
4.	A	11.	A	18.	C
5.	C	12.	C	19.	A
6.	B	13.	C	20.	A
7.	C	14.	C		